BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 040 · Ethical and Social Impacts

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why can defragmentation help a magnetic HDD?

Q2. Why use a different password for each account?
A. Make all passwords shorter
B. Remove the need for login
C. Give everyone equal access
D. Limit damage after a leak

Q3. How does encryption protect transmitted data?

Q4. A 20 × 10 image at 8 bits per pixel has 200 bytes of raw pixel data, but its uncompressed file is larger. Explain why this need not be an error and name two metadata examples.

I Do · Read new knowledge

Key vocabulary

Ethics — Judgements about what is fair, right or responsible.

Digital divide — Unequal access to digital devices, connectivity and skills.

Privacy — Control and protection of information about individuals.

E-waste — Discarded electrical or electronic equipment.

Cultural impact — How technology affects customs, language or social interaction.

Environmental impact — Effects on resources, energy, pollution and waste.

Accessibility — How readily people with different needs can use a service.

New learning

Ethics versus legislation: Ethical questions concern what is fair or responsible, even when an action is legal. Legal issues concern laws that apply. Consider the effect on different groups, not only technical capability.

Society and culture: Digital services make communication and collaboration easier, but unequal access to devices or reliable Internet creates a digital divide. Global digital communication can also affect language, culture and social interaction.

Environment: Electronic devices require energy and materials to produce, run and dispose of. Repair and recycling can reduce e-waste; increased energy demand has environmental costs.

Privacy and fair decisions: Data collection may improve services but can expose personal information. Biased automated decisions or inaccessible technology can disadvantage people, so design should be fair, transparent and inclusive.

Worked example

A town library introduces an online-only booking service. Explain TWO ethical or social concerns that the library should consider. [4 marks]

1. Identify the users: some visitors may have no broadband or suitable device.

2. Online-only booking could exclude those without digital access [1], preventing them from booking services easily [1].

3. Some visitors may lack digital skills or accessible interfaces [1], creating barriers to fair use [1].

4. Two concerns and linked effects. The library could also provide an in-person/telephone option.

materials.

Q2. Homework can be accessed anywhere with Internet [1], making resources easy to distribute/submit [1]. Pupils without reliable devices or connectivity [1] may find it harder to complete work, widening inequality [1].

Q3. Digital services can reduce travel/paper use [1], improving convenience and possibly resource use [1]. They may exclude people without reliable devices/skills [1], creating an access barrier [1]. Data collection could compromise privacy [1] without suitable protections, so a justified balance including alternative access is important [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It brings file fragments together, reducing read-head movement when accessing a file.

Q2. D — Limit damage after a leak

Q3. Encryption changes plaintext into ciphertext. Someone who intercepts it cannot easily read it without the key.

Q4. The file may include metadata or a header as well as pixels. Width, height and colour depth are examples. The formula counts only pixel data.

I Do / We Do pages · Guided practice

Q1. Patients who have smartphones can book appointments remotely [1], making access more convenient [1]. Patients without devices or with accessibility difficulties may be excluded [1], making healthcare harder to access [1].

Q2. Frequent replacement increases e-waste/material demand [1], creating disposal and manufacturing impacts [1]. Repair, reuse or recycle functioning tablets [1], reducing waste and demand for new materials [1].

Q3. Location data can enable more relevant nearby recommendations [1], helping users find suitable events [1]. Storing precise location histories [1] risks revealing where users visit if data is misused or exposed [1].

You Do pages · Code First

```
hasInternet = false
if hasInternet then
    print("Online form")
else
    print("Paper form")
endif
```

Expected result: Paper form

You Do pages · Independent questions

Q1. E-waste from discarded devices [1] and electricity/energy consumption [1]. Other clear valid concerns include manufacturing emissions or extracting raw

RHS COMPUTING / OCR J277 / 1.6.1 LESSON 040 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

We Do · Practise with guidance

Guided 1

A health centre makes appointments mainly through a mobile app. Explain two ways this could affect different patients. [4 marks]

Worked method: Patients who have smartphones can book appointments remotely [1], making access more convenient [1]. Patients without devices or with accessibility difficulties may be excluded [1], making healthcare harder to access [1].

Guided 2

A school replaces working tablets every two years. Explain an environmental concern and a possible mitigation. [4 marks]

Worked method: Frequent replacement increases e-waste/material demand [1], creating disposal and manufacturing impacts [1]. Repair, reuse or recycle functioning tablets [1], reducing waste and demand for new materials [1].

Guided 3

An online service collects location data to recommend local events. Explain one benefit and one privacy risk. [4 marks]

Worked method: Location data can enable more relevant nearby recommendations [1], helping users find suitable events [1]. Storing precise location histories [1] risks revealing where users visit if data is misused or exposed [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the access alternative displayed.

```
hasInternet = false
if hasInternet then
    print("Online form")
else
    print("Paper form")
endif
```

predict

Question 1

State TWO environmental concerns linked to digital technology. [2 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 2

A secondary school moves all homework online. Explain one benefit and one social disadvantage. [4 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 3

Evaluate how a council's move towards digital-only public services could affect citizens and the environment. [6 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

RHS COMPUTING / OCR J277 / 1.6.1 LESSON 040 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A service can be legal but still raise ethical concerns about fairness and access.